

Seat No.: _____

Stony Brook Institute at Anhui University

2022 Fall Semester

Fundamentals of Programming Final Exam (Paper A)

(Closed-book Exam Duration: 120 minutes)

Section	I	II	III	IV	Total score
Score					
Grader					

I. Multiple Choices (10 items, 1 point each)

Score

(Choose the best answer for each. The answers **must** be written in the form below.)

No.	1	2	3	4	5	6	7	8	9	10
Answer										

- Processing is a flexible software sketchbook and a programming language based on () for learning how to code.
A. Python B. C++ C. Java D. JavaScript
- Which of the following is not a data type of Processing?
A. integer B. double C. float D. boolean
- Which of the following assignment statements will not generate an error?
A. float a=2.0; B. char c="a"; C. byte b=2.5; D. boolean d=0;
- If x and y are double variables, the value of y is () after the statements x=1; y=x+3/2; are executed.
A. 1.0 B. 2.0 C. 3.0 D. 2.5
- Which of the following is a boolean expression for the relationship $x > y$ or $x \leq z$?
A. $(x > y) \& (x \leq z)$ B. $(x > y) \parallel (x \leq z)$
C. $(x > y) \mid (x \leq z)$ D. $(z \geq x > y)$
- When is the constructor of a class called?
A. When defining class B. When calling object methods
C. When creating object D. When using object variables
- Which of the following statements defining the String variable s is wrong?
A. String s = "metoo"; B. String s = new String("metoo");
C. String s; D. String s = new "metoo";
- Which of the following is the length of a String object?

- A. sizeof B. Length C. indexOf() D. length()
9. The rotate() function is identical to ().
 A. rotateX() B. rotateY() C. rotateZ() D. radians()
10. Which of the following is the default mode for the image() function?
 A. CORNER B. CORNERS C. CENTER D. CENTRE

II. Blank Fillings (10 items, 2 points each)

Score	
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(The answers **must** be written in the blanks below.)

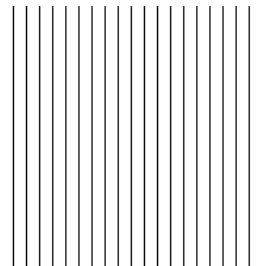
- (1) _____ (2) _____
 (3) _____ (4) _____
 (5) _____ (6) _____
 (7) _____ (8) _____
 (9) _____ (10) _____

1. // Finding the maximum number in the array.

```
int[] nums = {5, 4, 2, 7, 6, 8, 5, 2, 8, 14};
int max = nums[0];
for (int i = 1; _____ (1) _____; i++) {
    if (nums[i] > max) {
        _____ (2) _____;
    }
}
println(max);
```

2. // Drawing a series of vertical lines.

```
size(200, 200);
background(255);
for (int x=0; _____ (3) _____; x+=10) {
    stroke(0);
    _____ (4) _____;
}
```



3. // Rotating text

```
PFont f;
String message = "This text is spinning.";
float theta;
void setup() {
    size(200, 200);
```

```

    f = createFont("Arial", 18);
}
void draw() {
    background(255);
    fill(0);
    _____(5)_____;
    rotate(theta);
    textFont(f);
    textAlign(CENTER);
    _____(6)_____;
    theta += 0.01;
}

```

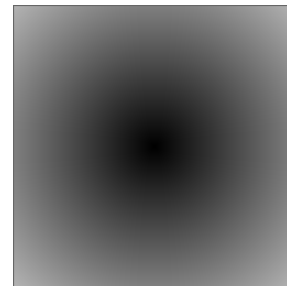


4. // Drawing the corresponding screenshot. The closer to the center of the display // window, the darker the color.

```

size(255, 255);
_____(7)_____;
for (int x = 0; x < width; x++) {
    for (int y = 0; y < height; y++) {
        int loc = _____(8)_____;
        float distance = dist(width/2, height/2, x, y);
        pixels[loc] = color(distance);
    }
}
updatePixels();

```



5. // Playing a video file.

```

import processing.video.*;
Movie movie;
void setup() {
    size(320, 240);
    movie = new Movie(this, "cat.mov");
    movie.play();
}
void _____(9)_____(Movie movie) {
    movie.read();
}
void draw() {
    _____(10)_____;
}

```

III. Corrections (5 items, 2 points each)

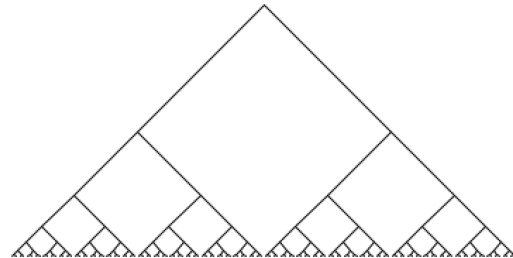
Score	
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- ```
1. float[] numbers = new float[10];
 for (int i = 0; i < numbers.length(); i++) {
 numbers[i] = random(2023);
 }
```
- ```
2. float z = 0;
   void setup() {
       size(200, 200);
   }
   void draw() {
       translate(width/2, height/2, z);
       rectMode(CENTER);
       rect(0, 0, 8, 8);
       z++;
   }
```
- ```
3. size(200, 200);
 PImage img = loadImage("flower.jpg");
 image(img, 0, 0, width, height);
```
- ```
4. class Car {
    int x, y;
    color c;
    float speed;
    void Car(int _x, int _y, int _c) {
        x = _x;
        y = _y;
        c = _c;
        speed = 0.0;
    }
}
```
- ```
5. Car[] cars = new Car(18);
 for (int i = 0; i < cars.length; i++) {
 cars[i] = new Car(0, i*10, color(random(255)));
 }
```

#### IV. Programming (5 items, 60 points in total)

| Score |  |
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1. Program to generate the following shapes (showed as the right-down figure) using recursion. (10 points)



2. Program to simulate a bouncing ball, which can bounce around in the horizontal direction. (10 points)

3. Program to display a 200×200 pixels image file named "sunflower.jpg" in the center of the display window, and rotate the image around its center. (10 points)

4. Write a Doorbell class which should change the color and play a doorbell sound whenever the user clicks on the circle. (15 points)

5. Using object-oriented programming to simulate a solar system, eight planets will orbit the sun. (15 points)